

Lecture 18

Saturday, April 18, 2026 9:24 AM

Recap: GDA (classification technique)

- Models $P(x|y)$, $P(y)$ (x : feature vector, y : class)
- Assumes $x|y=k \sim N(\mu_k, \Sigma)$, $y \sim \text{Ber}(\phi)$
- Parameters: $\phi, \mu_0, \mu_1, \Sigma$
- Uses MLE to estimate parameters (closed form, $O(Nd^2)$ complexity)
- Prediction: $\hat{y} = \underset{y}{\operatorname{argmax}} P(y|x) \leftarrow \underset{y}{\operatorname{argmax}} P(x|y) P(y)$ (Bayes' rule)

Naïve Bayes

Limitations of GDA

Spam filter (spam / not spam)

 $d = 50000$

set of 50000 "important" words: buy, free, offer, meeting, ...

$$x_j = \begin{cases} 1 & \text{if word } j \text{ occurs in the email} \\ 0 & \text{otherwise} \end{cases}$$

GDA on this:

$$\begin{array}{ll} \mu_k & k \in \{0, 1\} \rightarrow 50000 \quad (100,000) \\ \Sigma & \rightarrow 50000 \times 50000 : 25 \times 10^9 \end{array}$$

(Need something else to make this tractable)

- Naïve Bayes (conditional independence)It models $P(x|y)$ & $P(y)$

Assumes:

$$P(x_1, x_2, \dots, x_d | y) = \prod_{j=1}^d P(x_j | y)$$

Given the class, the features are independent

$d=3$ (say)

$$P(x_1, x_2, x_3 | y) = P(x_1 | y) P(x_2 | y) P(x_3 | y)$$

In reality:

$$P(x_1, x_2, x_3 | y) = P(x_1 | y) \underbrace{P(x_2 | y, x_1)}_{P(x_2 | y)} \underbrace{P(x_3 | y, x_1, x_2)}_{P(x_3 | y)}$$

$x_2 = \text{"free"}$
 $x_3 = \text{"offer"}$

$$P(\underline{x_2} | \underline{y}, \underline{x_3}) = P(\underline{x_2} | \underline{y}) \leftarrow \text{Naïve Bayes}$$

K class problem

$x_i \in \{0, 1\}$, d -dimensional features

$$\begin{array}{lll} P(y=k) & k=0, 1, \dots, K-1 & (K \text{ parameters}) \\ P(x_i | y=k) & d \times K & (dK \text{ parameters}) \end{array}$$

GDA : $O(d^2)$

Naïve Bayes : ~~$O(d^2)$~~ $(d+1)K$

Naïve Bayes Assumptions:

$$\textcircled{1} P(y=k) = \phi_k \quad \sum_{k=0}^K \phi_k = 1$$

$$\textcircled{2} P(x | y=k) = \prod_{j=1}^d P(x_j | y=k) \quad (\text{conditional independence})$$

$$\textcircled{3} x_j \in \{0, 1\}$$

$$x_j | y=k \sim \text{Ber}(\phi_{j|k})$$

$\downarrow$ class label
 $\downarrow$ feature index

Find $\phi_{j|k}$'s, ϕ_k 's $(d+1)K$ parameters

-① How to estimate the parameters? ✓

|② How to make a prediction for test samples?

↳ Training data: $\{x^{(i)}, y^{(i)}\}_{i=1}^N$

$$L = \prod_{i=1}^N P(x^{(i)}, y^{(i)}) \quad \text{--- (1)}$$

$$\ell = \log L = \sum_{i=1}^N \log P(x^{(i)}, y^{(i)}) \quad \text{--- (2)}$$

$$= \sum_{i=1}^N \log [P(x^{(i)} | y^{(i)}) P(y^{(i)})] \quad \text{--- (2) NB assumption}$$

$$= \sum_{i=1}^N \log \left[\prod_{j=1}^d P(x_j^{(i)} | y^{(i)}) P(y^{(i)}) \right] \quad \text{--- (4)}$$

$$= \sum_{i=1}^N \left(\underbrace{\log P(y^{(i)})}_{\text{class}} + \sum_{j=1}^d \underbrace{\log P(x_j^{(i)} | y^{(i)})}_{\text{features, class}} \right) \quad \text{--- (5)}$$

↑
training sample

Solution: (closed form, easy to interpret)

$$\textcircled{1} \phi_k^* = P(y = k) = \frac{\text{count}(y^{(i)} = k)}{N} \quad \left(\begin{array}{l} \text{fraction of} \\ \text{class } k \text{ samples} \\ \text{in the training data} \end{array} \right)$$

$k \in \{0, 1, \dots, K-1\}$

$$\textcircled{2} \phi_{j|k}^* = \frac{\text{count}(x_j^{(i)} = 1 \text{ and } y^{(i)} = k)}{\text{count}(y^{(i)} = k)} \quad \left(\begin{array}{l} k \in \{0, 1, \dots, K-1\} \\ j \in \{1, 2, \dots, d\} \end{array} \right)$$

Probability $x_j = 1$ given class is k

Going to back spam email example:

N - # of emails, ϕ_k^* = fraction of spam emails

j - feature index (corresponds to a word)

k - class index

$O(Nd)$ computation

Q. How to make a prediction?

$$\hat{y} = \underset{y}{\operatorname{argmax}} P(y = k | x)$$

$\vec{v}$
feature vector of
test sample

$$\begin{aligned}
 &= \operatorname{argmax}_y P(x|y=k) P(y=k) \\
 &= \operatorname{argmax}_y \prod_{j=1}^d \underbrace{P(x_j|y=k)}_{=\phi_{j|k}^*} \underbrace{P(y=k)}_{=\phi_k^*}
 \end{aligned}$$

We saw an issue with multiplications by 0.
We need a fix.

Laplace Smoothing

- We pretend we saw every word at least once in each class.

$$\begin{aligned}
 \hat{\phi}_{j|k} &= \frac{\text{count}(x_j=1 \text{ \& class } k)}{\text{count}(\text{class } k)} \quad \left. \begin{array}{l} \text{original} \end{array} \right\} \times \\
 &= \frac{\text{count}(x_j=1 \text{ \& class } k) + 1}{\text{count}(\text{class}) + \text{number of values } x_j \text{ can take}} \quad \left. \begin{array}{l} \text{new} \end{array} \right\} \checkmark
 \end{aligned}$$

$$\hat{\phi}_{j|k}(\text{new}) > 0$$

Question: $\searrow$ independence

Case I: $A \perp\!\!\!\perp B \Rightarrow P(A \cap B) = P(A) P(B)$

$\searrow$ Case II: $A \perp\!\!\!\perp B | C \Rightarrow P(A \cap B | C) = P(A | C) P(B | C)$
 $\nearrow$ conditionally independence

All models are wrong; but some are useful

